

Physics-0 2026 Spring

Homework 10

程昊一

*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1. (20pts) An oxygen-16 ion with a mass of 2.66×10^{-26} kg travels at 5×10^6 m/s perpendicular to a 1.20 T magnetic field, which makes it move in a circular arc with a 0.231 m radius. What is the ratio of the charge of the ion to the charge of the electron? Does this result make sense?

Solution. Suppose the charge of the ion is q . The magnetic force on the ion is given by

$$|\mathbf{F}| = qvB = m \frac{v^2}{R} \implies q = \frac{mv}{RB}.$$

where m is the mass of the ion, v is its velocity, R is the radius of the circular arc, and B is the magnetic field strength. Substituting the given values, we find

$$q = \frac{2.66 \times 10^{-26} \text{ kg} \times 5 \times 10^6 \text{ m/s}}{0.231 \text{ m} \times 1.20 \text{ T}} = 4.798 \times 10^{-19} \text{ C}.$$

The charge of an electron is approximately -1.602×10^{-19} C. Therefore, the ratio of the charge of the ion to the charge of the electron is approximately -3 . This result indicates that the ion is O^{3+} .

Problem 2. (40pts) Consider a square loop of length $4L$ with current I and magnetic dipole moment μ . The magnetic dipole moment makes an angle θ_1 with the positive z -axis.

- (a) (5pts) Does the angle θ_1 increase or decrease when a uniform magnetic field $\mathbf{B}$ pointing in the $\hat{z}$ direction is turned on?
- (b) (10pts) Compute the work done by the magnetic field $\mathbf{B}$ as the loop rotates from θ_1 to θ_2 .
- (c) (20pts) Use the expression above to find an expression for the potential energy of the loop in terms of the magnetic dipole moment.
- (d) (5pts) How does this expression compare to the potential energy of an electric dipole in the presence of a uniform electric field?

Solution. (a) The torque τ acts on the magnetic dipole is given by

$$\tau = \mu \times \mathbf{B}.$$

Then the angle between μ and $\mathbf{B}$ will decrease when the magnetic field is turned on, so the angle θ_1 will decrease.

(b)

$$W = \int_{\theta_1}^{\theta_2} \tau(\theta) d\theta = - \int_{\theta_1}^{\theta_2} \mu B \sin \theta d\theta = \mu B (\cos \theta_2 - \cos \theta_1).$$

(c) We can find a potential energy of a magnetic dipole in a magnetic field as

$$U = -\boldsymbol{\mu} \cdot \mathbf{B}.$$

The relation $\Delta U = -W$ can be easily verified.

(d) The potential energy of an electric dipole $\mathbf{p}$ in a uniform electric field $\mathbf{E}$ is given by

$$U = -\mathbf{p} \cdot \mathbf{E}.$$

This expression is remarkably similar to the potential energy of a magnetic dipole in a magnetic field.

Problem 3. (40pts) Consider a square loop on the xy -plane with current I flowing clockwise. The corners of the loop are located at $(0, 0)$, $(0, L)$, $(L, 0)$, and (L, L) . There is a nonuniform magnetic field that is given by

$$\mathbf{B} = \frac{B_0 z}{L} \hat{y} + \frac{B_0 y}{L} \hat{z},$$

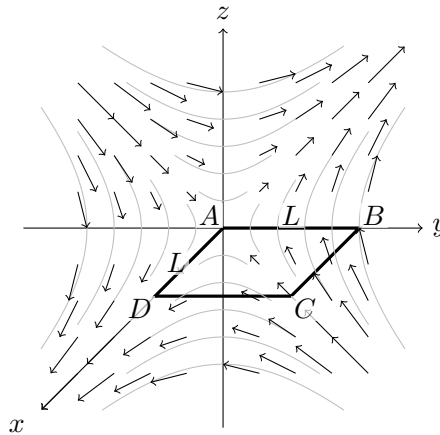
where B_0 is a positive constant.

(a) (10pts) Sketch the magnetic field lines in the yz -plane.

(b) (20pts) Find the magnitude and direction of the magnetic force exerted on each side of the loop.

(c) (10pts) Find the magnitude and direction of the net magnetic force on the loop.

Solution. (a) The sketch of the magnetic field is shown in the figure below. To show the field more clearly, the field lines are also plotted.



(b)

$$\begin{aligned} \mathbf{F}_{AB} &= \int_0^L I d\mathbf{l} \times \mathbf{B} = \int_0^L I \frac{B_0 l}{L} d\mathbf{l} \hat{x} = \frac{1}{2} I B_0 L \hat{x}. \\ \mathbf{F}_{BC} &= \int_0^L I d\mathbf{l} \times \mathbf{B} = \int_0^L I d\mathbf{l} \hat{x} \times B \hat{z} = -I B_0 L \hat{y}. \\ \mathbf{F}_{CD} &= \int_L^0 I d\mathbf{l} \times \mathbf{B} = \int_L^0 I \frac{B_0 l}{L} d\mathbf{l} \hat{x} = -\frac{1}{2} I B_0 L \hat{x}. \\ \mathbf{F}_{DA} &= \int_L^0 I d\mathbf{l} \times \mathbf{B} = 0. \end{aligned}$$

(c)

$$\mathbf{F} = \mathbf{F}_{AB} + \mathbf{F}_{BC} + \mathbf{F}_{CD} + \mathbf{F}_{DA} = -I B_0 L \hat{y}.$$